Time: 1-4 PM

Name:

Student ID:

Attached are two sections of questions.

Theoretical questions (25)

1. How does CIR differ from traditional Web search?
2. What is the purpose of query rewriting in conversational systems?
3. Define the term “belief state” in dialogue systems.
4. Describe the architecture of a C-KBQA system and its key components.
5. Why is tracking dialog state important in multi-turn QA?
6. Compare extractive QA and abstractive QA in the context of CIR.
7. Discuss the limitations of BERT in multi-turn QA and how FlowQA addresses them.
8. Describe how adversarial attacks can affect MRC models like BERT. Give an example.

Practical questions (25)

Choose one short paragraph (5–6 sentences) or a simple table with 3–4 rows.

Write three related questions in sequence, where each question depends on the answer to the previous one (as in CoQA, SQA, or QuAC).

Implement the system using one of the following models:

bert-large-uncased-whole-word-masking-finetuned-squad

google/tapas-large-finetuned-wtq (if using table)

BiDAF++ or simulation of FlowQA with LSTM/Transformer (optional for advanced students)

For each question, run:

Without conversation history

With prepended or rewritten history

Compare the model's answers in both cases.

Add one **adversarial sentence** to the context (e.g., misleading or confusing detail) and re-evaluate the answer.

Explain the following:

Did history improve answer accuracy?

Was the model affected by the adversarial input?

Required

- Python code or Jupyter Notebook (`.ipynb`)
- Printed answers for all 3 questions (with and without history)
- Print of adversarial result
- A short report (half a page) that includes:
  - Model used and reason for choosing it
  - Observations about history handling
  - Limitations observed in the model

Allowed Libraries:

- transformers (HuggingFace)
- datasets
- torch
- pandas (if using tables)
- Optional: allennlp for BiDAF++ (advanced)

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